Constructs a `std::mutex` instance.

Postconditions

The newly constructed `std::mutex` object is initially unlocked.

Throws

Nothing.

Std::Mutex Destructor

Destroys a `std::mutex` object.

Declaration

```
~mutex();
```

Preconditions

`*this` must not be locked.

Effects

Destroys `*this`.

Throws

Nothing.

Std::Mutex::Lock Member Function

Acquires a lock on a `std::mutex` object for the current thread.

Declaration

```
void lock();
```